

WitnessKey / LOOPtLOOP / Abracadoo

Private Authorization Witnessing v0.1 - Developer Implementation Spec

Canonical API destination: <https://api.looptloop.online/v0>

Field	Value
Date	2026-06-19
Version	0.1 updated domain note
Status	Draft for first implementation
Primary service wedge	Private Authorization Witnessing
Intended API host	api.looptloop.online , prepared and ready on Cloudflare
Core invariant	WitnessKey never needs to see the private authorization payload. It witnesses only a hash, consent envelope, declared roles, timing window, schema/code version, and receipt signature.

0. Domain Update / Deployment Intention

Use <https://api.looptloop.online/v0> as the intended base URL for the LOOPtLOOP API service. The prior placeholder api.looptloop.io should be treated as deprecated because looptloop.io has lapsed. The currently held and Cloudflare-prepped domain is **looptloop.online**, with the API destination intended to be **api.looptloop.online**.

Canonical base URL:

<https://api.looptloop.online/v0>

Example endpoint:

POST <https://api.looptloop.online/v0/authorization-offers>

1. Purpose

This specification defines the first working implementation of a privacy-preserving authorization witness service. It allows two or more participants - such as a user and an AI agent, a user and an app, two users, or any combination of loop-capable entities - to privately exchange or agree to an authorization payload. WitnessKey / LOOPtLOOP then witnesses only the hash and metadata envelope of that private authorization event.

The result is an auditable third-party receipt confirming that an authorization event was witnessed, without requiring the witness service to receive or store the sensitive authorization content itself.

2. Product Summary

- One-sentence positioning: WitnessKey provides privacy-preserving third-party confirmation of authorization events by witnessing hashes and consent envelopes instead of private payloads.
- Compressed architecture: Abracadoo participates. WitnessKey witnesses. LOOPtLOOP binds.
- Primary v0.1 event type: `private_authorization_witnessed`.

System	Role
WitnessKey / WitnessMark	User-facing witness surface, receipt display, verification page, hash-only event witness UI.
LOOPtLOOP API	Consent loop / witness event API, offer lifecycle, event record, receipt signature, verification endpoint. Hosted at api.looptloop.online .

Abracadoo.app	Participant runtime, consent acceptance interface, local receipt storage, local path/loop-aware wallet.
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3. v0.1 User Flow

1. User opens WitnessKey / WitnessMark.
2. User enters or generates a private authorization payload locally.
3. Browser computes payload_hash.
4. User consents to hash-witnessing.
5. WitnessKey creates a witness offer through the LOOPtLOOP API at <https://api.looptloop.online/v0>.
6. User is sent to Abracadoo to review and accept the offer.
7. Abracadoo displays the issuer, hash, declared roles, storage policy, and consent prompt.
8. User accepts.
9. Abracadoo posts acceptance to LOOPtLOOP.
10. LOOPtLOOP creates a witness event and signed receipt.
11. Abracadoo stores the receipt locally.
12. User returns to WitnessKey.
13. WitnessKey displays the receipt and verification link.

4. Privacy Property

The private authorization payload remains local/private unless a participant later chooses to reveal it. WitnessKey stores only event metadata required to verify the authorization witness receipt.

WitnessKey stores	WitnessKey does not store
payload hash	private authorization payload
schema version	raw conversation
role declarations	browser history
consent prompt hash	legal identity
storage policy	agent private reasoning
timestamp/window	unrelated activity
receipt signature	
optional participant public key or pseudonymous reference	

5. Core Schemas

5.1 Authorization Offer

```
{
  "schema": "WITNESSKEY_AUTHORIZATION_OFFER_0_1",
  "offer_id": "wo_01JXYZ...",
  "event_type": "private_authorization_witnessed",
  "issuer": {
    "name": "WitnessKey",
    "origin": "https://witnesskey.online"
  },
  "subject": {
    "type": "private_payload_hash",
    "hash_alg": "SHA-256",
    "payload_hash": "sha256:..."
  }
}
```

```

},
"payload_label": "AI agent authorization",
"declared_roles": ["human_authorizer", "ai_agent_actor"],
"consent_prompt": "Do you consent to witness this private authorization by hash?",
"created_at": "2026-06-19T00:00:00.000Z",
"expires_at": "2026-06-19T00:05:00.000Z",
"return_url": "https://witnesskey.online/receipt-pending?offer_id=wo_01JXYZ..."
}

```

5.2 Authorization Event

```

{
  "schema": "WITNESSKEY_AUTHORIZATION_EVENT_0_1",
  "event_id": "we_01JXYZ...",
  "offer_id": "wo_01JXYZ...",
  "loop_id": "loop_01JXYZ...",
  "event_type": "private_authorization_witnessed",
  "participants": [
    {"role": "client_participant", "app": "abracadoo.app"},
    {"role": "server_witness", "app": "witnesskey.online"}
  ],
  "subject": {
    "type": "private_payload_hash",
    "hash_alg": "SHA-256",
    "payload_hash": "sha256:..."
  },
  "created_at": "2026-06-19T00:01:30.000Z",
  "expires_at": "2026-06-19T00:05:00.000Z",
  "receipt_signature": "base64url-signature-or-dev-hmac",
  "verification_url": "https://witnesskey.online/verify/we_01JXYZ..."
}

```

6. API Endpoints

Canonical base URL:

<https://api.looptloop.online/v0>

Method	Path	Purpose
POST	/authorization-offers	Create a short-lived offer to witness a private authorization payload by hash.
GET	/authorization-offers/:offer_id	Fetch offer details for Abracadoo review.
POST	/authorization-offers/:offer_id/accept	Accept offer after explicit Abracadoo-side consent.
POST	/authorization-offers/:offer_id/reject	Reject offer; no witness event is created.
GET	/authorization-events/:event_id	Return public, non-sensitive event metadata.
GET	/authorization-events/:event_id/verify	Return verification status and receipt summary.

7. Endpoint Examples

7.1 Create Authorization Offer

POST <https://api.looptloop.online/v0/authorization-offers>

Request:

```
{
  "schema": "WITNESSKEY_AUTHORIZATION_OFFER_CREATE_0_1",
  "issuer": {"name": "WitnessKey", "origin": "https://witnesskey.online"},
  "event_type": "private_authorization_witnessed",
  "payload_hash": "sha256:...",
  "payload_label": "AI agent authorization",
  "declared_roles": ["human_authorizer", "ai_agent_actor"],
  "consent_prompt": "Do you consent to witness this private authorization by hash?",
  "return_url": "https://witnesskey.online/receipt-pending"
}
```

Response:

```
{
  "offer_id": "wo_01JXYZ...",
  "status": "offered",
  "accept_url": "https://abracadoo.app/accept-witness?offer_id=wo_01JXYZ...",
  "expires_at": "2026-06-19T00:05:00.000Z"
}
```

7.2 Accept Authorization Offer

POST https://api.looptloop.online/v0/authorization-offers/:offer_id/accept

Request:

```
{
  "schema": "WITNESSKEY_AUTHORIZATION_ACCEPTANCE_0_1",
  "accepted_by": {
    "app": "abracadoo.app",
    "participant_ref": "local:pseudonymous-path-id-or-public-key",
    "participant_role": "human_authorizer"
  },
  "consent_action": "accept",
  "consent_prompt_hash": "sha256:...",
  "participant_signature": "optional-v0.1"
}
```

8. Database Schema - Cloudflare D1

```
CREATE TABLE authorization_offers (
  id TEXT PRIMARY KEY,
  schema_version TEXT NOT NULL,
```

```

event_type TEXT NOT NULL,
issuer_name TEXT NOT NULL,
issuer_origin TEXT NOT NULL,
payload_hash TEXT NOT NULL,
payload_label TEXT,
declared_roles_json TEXT NOT NULL,
consent_prompt TEXT NOT NULL,
consent_prompt_hash TEXT NOT NULL,
storage_policy_json TEXT NOT NULL,
claims_policy_json TEXT NOT NULL,
return_url TEXT,
created_at TEXT NOT NULL,
expires_at TEXT NOT NULL,
status TEXT NOT NULL
);

```

```

CREATE TABLE authorization_events (
  id TEXT PRIMARY KEY,
  offer_id TEXT NOT NULL,
  loop_id TEXT NOT NULL,
  schema_version TEXT NOT NULL,
  event_type TEXT NOT NULL,
  issuer_name TEXT NOT NULL,
  issuer_origin TEXT NOT NULL,
  payload_hash TEXT NOT NULL,
  participant_app TEXT,
  participant_ref TEXT,
  participant_role TEXT,
  receipt_json TEXT NOT NULL,
  receipt_signature TEXT NOT NULL,
  verification_url TEXT NOT NULL,
  created_at TEXT NOT NULL,
  expires_at TEXT,
  status TEXT NOT NULL,
  FOREIGN KEY(offer_id) REFERENCES authorization_offers(id)
);

```

9. WitnessKey / WitnessMark Client Changes

- Rename external mode to: External WitnessKey authorization loop.
- Replace the placeholder external function with offer creation, status polling, and receipt rendering.
- In external mode, compute payload_hash locally and send only the hash/envelope to <https://api.looptloop.online/v0>.
- Keep existing consent gates before creating an external witness offer.
- Render the returned receipt and preserve copy/download/print behavior.

10. Abracadoo Client Changes

- Add route: /accept-witness?offer_id=...
- Fetch offer from https://api.looptloop.online/v0/authorization-offers/:offer_id.
- Display issuer, event type, payload hash, payload label, declared roles, storage policy, non-storage policy, claims, non-claims, and expiration.
- Require explicit user acceptance before posting to the accept endpoint.
- Store returned receipt locally. Do not store private authorization payload unless the user explicitly chooses to save it locally.
- Show: This creates a witnessed authorization event. It does not automatically create a Relationship.

11. Offer Lifecycle

Offer statuses:

offered -> accepted | rejected | expired | revoked | error

Event statuses:

active -> expired | revoked | superseded | error

- Offers expire after five minutes by default.
- Expired offers cannot be accepted.
- Accepted offers cannot be accepted again.
- Rejecting an offer terminates it and creates no event.

12. Security and Privacy Requirements

12.1 Must

- Never require private payload upload.
- Never store private payload by default.
- Expire offers.
- Prevent accepting expired offers.
- Prevent double acceptance.
- Record claims and non-claims.
- Make storage policy visible before acceptance.
- Make verification possible without revealing private payload.
- Rate-limit public offer creation.
- Use CORS allowlists for known app origins during v0.1.

12.2 Must not

- Claim legal consent.
- Claim legal identity.
- Claim authorship.
- Claim truth.
- Claim that the authorized action was actually performed.
- Claim absence of coercion.
- Store hidden behavioral history.
- Use the private payload for analytics.

13. Non-Goals for v0.1

- accounts

- billing
- organizations
- OAuth
- Web3 wallet integration
- NFT logic
- hardware/NFC/ESP32 anchors
- healthcare compliance mode
- legal identity verification
- multi-party loops
- AI-agent automation execution
- production-grade key management
- mobile push notifications
- full loop directory

14. First Test Cases

- Create offer: valid payload hash and consent prompt returns offer ID and Abracadoo accept URL.
- Fetch offer: Abracadoo displays correct payload hash and storage policy.
- Accept offer: unexpired offer returns witness event and receipt.
- Reject offer: no event is created.
- Expired offer: acceptance fails.
- Verify event: verification endpoint returns signature status and public metadata.
- Private payload never stored: database contains payload_hash but not raw private text.
- Receipt comparison: original private payload can be hashed locally and compared to receipt payload hash.

15. First Sprint Deliverables

- JSON schemas for offer, acceptance, event, and receipt.
- Cloudflare Worker API deployed at <https://api.looptloop.online/v0>.
- D1 database migration.
- WitnessMark external mode connected to API.
- Abracadoo accept-witness route.
- Abracadoo local receipt storage.
- WitnessKey verification page.
- One complete demo receipt.

16. Canonical Invariant

Private payload stays private.

WitnessKey witnesses the hash, consent envelope, declared roles, timing window, schema/code version, and receipt signature.

End of spec.